Prodigy Anesthesia Quick Review

Tuberculosis

What is the organism responsible for tuberculosis?	Mycobacterium tuberculosis *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 481.*
Why does the organism responsible for tuberculosis infect the apices of the lungs preferentially?	The organsism responsible for tuberculosis (Mycobacterium tuberculosis) requires high oxygen concentrations to thrive. Because of the natural tendency of oxygen to maintain higher concentrations in the superior portion of the lung, tuberculosis has an increased incidence in the apices. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 481.*
What are the most common symptoms of tuberculosis?	Nonproductive cough, hemoptysis, night sweats, chest pain, and weight loss. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 481-482.*
What types of patients exhibit the highest incidence of tuberculosis?	Foreign-born persons from areas where the disease is prevalent (Africa and Asia), intravenous drug abusers, and patients with AIDS. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 481-482.*
What chest ray findings are associated with tuberculosis?	Tuberculosis is associated with chest xray findings such as bilateral upper lobe infiltrates (often with cavitation), apical and subapical infiltrates, and vertebral osteomyelitis (Pott's disease) which is common in advanced extrapulmonary tuberculosis. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 482.*
What airway and respiratory considerations should you be aware of when caring for a patient with tuberculosis?	If tuberculosis of the cervical spine is present, airway management should proceed cautiously to prevent injury. Patients with tuberculosis need to have a high efficiency airborne particulate filter placed between the patient and the anesthesia circuit. During transport, the patient must wear a N-95 filter mask. During surgical procedures, all healthcare workers in the operating room must also wear N-95 masks. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 482.*
Rifampin is a drug used in the treatment of tuberculosis. What are some of the serious side effects of this drug?	Thrombocytopenia, leukopenia, anemia, renal failure, and hepatic failure. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 482.*

Copyright 2015 Prodigy Anesthesia, Inc. Unlawful to duplicate, distribute, or sell this document.

Prodigy Anesthesia Quick Review

Tuberculosis

Isoniazid is used in the treatment of tuberculosis. What are some of the more serious side effects of this drug?	Isoniazid is toxic and can result in damage to the liver, peripheral nervous system, and kidneys. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 482.*
What is the most common test used to diagnose tuberculosis and what are its limitations? What tests are more reliable for diagnosing TB?	The tuberculin skin test (Mantoux test) is the most common test used to diagnose TB. The test may inaccurately report positive if the patient has been exposed but not infected with TB, has received a bacille Calmette-Guerin vaccine, or been exposed to other species of Mycobacterium, even if there is no bacterial infection at the time of testing. Chest radiographs and sputum smears are more accurate in diagnosing TB. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 481.*
How effective is current medical therapy in the treatment of tuberculosis?	Although multi-drug resistant strains of tuberculosis are beginning to result in a resurgence in the incidence of TB, current medical therapy results in a negative sputum smear in 90% of patients within 3 months. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 481.*
A patient with old pulmonary tuberculosis is about to undergo general anesthesia. What concerns do you have regarding spontaneous versus positive-pressure ventilation?	Positive-pressure ventilation can result in massive hemoptysis in patients with long-standing pulmonary TB. Many practitioners prefer to employ spontaneous ventilation if at all possible to reduce the risk of hemoptysis. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 482.*

Copyright 2015 Prodigy Anesthesia, Inc. Unlawful to duplicate, distribute, or sell this document.